General Notes on FEOM

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This PDF contains general notes on the FEOM project, including information about the project structure, coding conventions, and other relevant details.

I. Project Structure

This project implements HEOM using FORTRAN, and implements basic HEOM with the standard K and L cuttoffs. In the notation used by FORTRAN propagation, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_{\boldsymbol{n}}(t) = -\frac{i}{\hbar}H_s^{\times}\rho_{\boldsymbol{n}}(t) - \sum_k n_k\gamma_k\rho_{\boldsymbol{n}}(t) - \frac{i}{\hbar}\sum_k \left(c_{\mathrm{U}}(k,n_k)V^{\times}\rho_{\boldsymbol{n}_k^+}(t)\right)$$
$$+\frac{i}{\hbar}\sum_k\left(c_{\mathrm{D_{LEFT}}}(k,n_k)V^L\rho_{\boldsymbol{n}_k^-}(t) + c_{\mathrm{D_{RIGHT}}}(k,n_k)V^R\rho_{\boldsymbol{n}_k^-}(t)\right), \tag{1}$$

where $\rho_{\boldsymbol{n}}(t)$ is the full set of ADOs, $\boldsymbol{n}$ is the vector of ADO indices $\{n_k\}$, (with $\boldsymbol{n}_k^{\pm}$ denoting the same set of indices, but with $n_k \pm 1$). We use superoperator notation throughout, and define our coefficients $\{\gamma_k, c_{\mathrm{U}}, c_{\mathrm{D}}\}$ differently for each version of the HEOM used.

A. Shi/Chen scaled HEOM

For Shi/Chen scaled HEOM, the equations of motion are given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_{\boldsymbol{n}}(t) = -\frac{i}{\hbar}H_s^{\times}\rho_{\boldsymbol{n}}(t) - \sum_k n_k\gamma_k\rho_{\boldsymbol{n}}(t) - \frac{i}{\hbar}\sum_k\left(\sqrt{n_k+1}\mathcal{L}_{k-}\rho_{\boldsymbol{n}_k^+}(t) + \sqrt{n_k}\mathcal{L}_{k+}\rho_{\boldsymbol{n}_k^-}(t)\right), \tag{2}$$

where $\rho_{\boldsymbol{n}}(t)$ is the full set of ADOs, $\boldsymbol{n}$ is the vector of ADO indices $\{n_k\}$, (with $\boldsymbol{n}_k^{\pm}$ denoting the same set of indices, but with $n_k \pm 1$). We use superoperator notation throughout, H_s as the system Hamiltonian, the superoperators $\mathcal{L}_{k-}$ and $\mathcal{L}_{k+}$ are defined as the superoperator

$$\mathcal{L}_{k-}\cdot = \sqrt{|C_k|}V^{\times}, \qquad \mathcal{L}_{k+}\cdot = \frac{1}{\sqrt{|C_k|}}\left(C_kV^L - C_k^*V^R\right), \tag{3}$$

where V is the system-bath coupling operator, C_k is the coupling strength for the k'th bath mode, and the constants C_k and γ_k are obtained from the bath correlation function (BCF) as

$$C(t) = \sum_k C_k e^{-\gamma_k t}, \tag{4}$$

which results in the coefficients,

$$c_{\mathrm{U}}(k,n_k) = \sqrt{n_k+1}\,|C_k|\,, \quad c_{\mathrm{D_{LEFT}}}(k,n_k) = -\sqrt{\frac{n_k}{|C_k|}}C_k, \quad c_{\mathrm{D_{RIGHT}}}(k,n_k) = \sqrt{\frac{n_k}{|C_k|}}C_k^*, \tag{5}$$

and γ_k being the decay rate of the k'th bath mode, and K modes chosen to represent the BCF. In what follows, any sums over k are assumed to go from 1 to K, unless otherwise specified.

B. Free-Pole HEOM

For Free-Pole HEOM, the bath correlation function is represented in terms of a sum of dampoed exponentials,

$$C_k = \sum_{k=1}^{K} d_k e^{-z_k t}, \tag{6}$$

with d_k , $z_k \in \mathbb{C}$. the equations of motion are given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\rho_{\boldsymbol{m},\boldsymbol{l}}(t) = -\frac{i}{\hbar}H_s^{\times}\rho_{\boldsymbol{m},\boldsymbol{l}}(t) - \left(\sum_k m_k z_k + l_k z_k^*\right)\rho_{\boldsymbol{m},\boldsymbol{l}}(t) - \frac{i}{\hbar}\sum_k \sqrt{(m_k+1)d_k}V^{\times}\rho_{\boldsymbol{m}_k^+,\boldsymbol{l}}(t)$$
$$-\frac{i}{\hbar}\sum_k \sqrt{(l_k+1)d_k^*}V^{\times}\rho_{\boldsymbol{m},\boldsymbol{l}_k^+}(t) - \frac{i}{\hbar}\sum_k \sqrt{m_k d_k}V^{L}\rho_{\boldsymbol{m}_k^-,\boldsymbol{l}}(t) + \frac{i}{\hbar}\sum_k \sqrt{l_k d_k^*}V^{R}\rho_{\boldsymbol{m},\boldsymbol{l}_k^-}(t). \quad (7)$$

We can combine the two ADO intices (there are two of them due to the forward and back paths being separated out in the influence functional) into a single vector $\boldsymbol{n} = \{\boldsymbol{m}, \boldsymbol{l}\}$, such that the EOM are now in the same form as the Shi/Chen scaled HEOM, with the coefficients defined as

$$c_{\mathrm{U}}(k, n_k) = \begin{cases} \sqrt{(n_k+1)d_k}, & k \le K, \\ \sqrt{(n_k+1)d_{k-K}^*}, & k > K \end{cases}, \qquad c_{\mathrm{D_{LEFT}}}(k, n_k) = \begin{cases} -\sqrt{n_k d_k}, & k \le K, \\ 0, & k > K \end{cases} \quad (8)$$

$$c_{\mathrm{D_{RIGHT}}}(k, n_k) = \begin{cases} 0, & k \le K, \\ \sqrt{n_k d_{k-K}^*}, & k > K \end{cases}, \qquad \gamma_k = \begin{cases} z_k, & k \le K, \\ z_{k-K}^*, & k > K \end{cases}. \quad (9)$$

II. Bath conventions

In this section, we discuss the conventions used for the naming of the baths. We define the spectral density

$$J(\omega) = \frac{\pi}{2}\sum_\alpha \frac{c_\alpha^2}{m_\alpha \omega}\left[\delta(\omega-\omega_\alpha) + \delta(\omega+\omega_\alpha)\right], \quad (10)$$

where c_α is the coupling strength, m_α is the mass, and ω_α is the frequency of the α'th bath mode. It is assumed that $\omega_\alpha \ge 0$, such that $J(\omega) = -J(-\omega)$. With this definition, the equilibrium bath correlation function (BCF), at inverse temperature β, is given as

$$C(t) = \frac{\hbar}{2\pi}\int_{-\infty}^{\infty} d\omega J(\omega)\left(\coth\left(\frac{\beta\hbar\omega}{2}\right)\cos\omega t - i\sin\omega t\right), \quad (11)$$

which is done using residues. Now to get onto specific bath distributions.

A. Debye bath

The Debye bath is defined by the spectral density

$$J(\omega) = \frac{\eta\gamma\omega}{\omega^2+\gamma^2}, \quad (12)$$

where η is the coupling strength and γ is the cutoff frequency. The correlation function is of exponential form, paramaterized by frequencies $\{\gamma_k\}$,

$$\gamma_0 = \gamma, \qquad \gamma_k = \omega_{n=k} = \frac{2n\pi}{\beta\hbar}, \quad (13)$$

and prefactors $\{C_k\}$, which are given by

$$C_0 = \frac{\hbar\eta\gamma}{2}\left(\cot\frac{\beta\hbar\gamma}{2} - i\right), \qquad C_{n=k} = -\frac{2\eta\gamma}{\beta}\frac{\omega_n}{\gamma^2-\omega_n^2}, \qquad \forall k > 0. \quad (14)$$

III. Truncations

There are a couple ways to truncate the HEOM, so it's good to write these down. The truncation falls into two categories, which I will call exponential term truncation and heirarchy truncation. The exponential term truncation is the approximation of the bath correlation function (BCF) by a finite number of exponentials. The heirarchy truncation is the removal of the dependence of the ADOs on those with a larger depth.

A. Matsubara series truncation

This truncation was created by Ishizaki and Tanimura in Ref.[1],and is switched on by the variable "*lowTcoef_switch*". It is a Markovian approximation of the higher order Matsubara terms in the BCF;

$$e^{-\gamma_k t} \approx \frac{1}{\gamma_k}\delta_+(t), \qquad \forall k > k_c, \tag{15}$$

where k_c is the cutoff in number of exponentials in the BCF[1] and $\delta_+(t)$ is a delta function defined for $t \geq 0$. This cutoff $k_v = M + 1$ for the example of a Debye bath, wherein there is a single exponential term, and M is the number of Matsubara terms retained. This approximation is inserted into the influence functional, allowing us to do the inner integral (over the history), giving another term in the dynamics,

$$\Xi\rho_{\boldsymbol{n}}(t) = \sum_{k>k_c} \frac{1}{\gamma_k}\mathcal{L}_{k-}\mathcal{L}_{k+}\rho_{\boldsymbol{n}}(t). \tag{16}$$

For the example of a Debye bath, we can do the sum formally. Due to all of the C_k for $k > 0$ being real, we can write the superoperator as

$$\Xi_{\text{debye}} = -\frac{1}{\hbar^2}\sum_{k>k_c}\frac{1}{\gamma_k}C_k\left(V^\times\right)^2 = \frac{2\eta\gamma}{\beta\hbar^2}\sum_{n>k_c}\frac{1}{\gamma^2-\omega_n^2}\left(V^\times\right)^2 \tag{17}$$

Then we use the fact that

$$\cot z = \frac{1}{z} + \sum_{n=1}^{\infty}\frac{2z}{z^2-n^2\pi^2}$$

$$\therefore \sum_{n=1}^{\infty}\frac{1}{\gamma^2-\omega_n^2} = \frac{\beta\hbar}{4\gamma}\left(\cot\frac{\beta\hbar\gamma}{2} - \frac{2}{\beta\hbar\gamma}\right), \tag{18}$$

to calculate the sum, giving us

$$\Xi_{\text{debye}} = \underbrace{\eta\left[\frac{1}{2\hbar}\left(\cot\frac{\beta\hbar\gamma}{2} - \frac{2}{\beta\hbar\gamma}\right) - \frac{2\gamma}{\beta\hbar^2}\sum_{n=1}^{k_c}\frac{1}{\gamma^2-\omega_n^2}\right]}_{lowTcoef}\left(V^\times\right)^2. \tag{19}$$

References

[1] Akihito Ishizaki and Yoshitaka Tanimura. Quantum Dynamics of System Strongly Coupled to Low-Temperature Colored Noise Bath: Reduced Hierarchy Equations Approach. *Journal of the Physical Society of Japan*, 74(12):3131–3134, December 2005.

[1] these are ordered in increasing real part, as then the approximation is more accurate for larger k